

# Future Service Rights: Tokenization, Pricing, and Structuring

Theory and Evidence from the Chengdu Hotel Market

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May 2026

## Abstract

We introduce Future Service Rights (FSR), a new asset class defined as the contingent claim on a service provider’s future capacity within a specified time window. Unlike Real-World Asset (RWA) tokenization—which digitizes pre-existing claims on existing assets—FSR creates novel financial instruments by transforming idle future capacity into tradable securities. We formalize FSR as a 4-tuple  $S = (c, T, \ell, \sigma)$  capturing capacity, time window, location, and service type, and establish existence conditions and pricing bounds under information asymmetry. As an instantiation, we develop Time-Rights for the hotel sector: tokenized forward room-night claims that trade on secondary markets and settle via a ternary mechanism. Critically, we extend this mechanism with a **continuous secondary-market offset**: Time-Right holders can sell their positions at any time before maturity, with the proceeds directly applied as booking credit toward any hotel reservation. This transforms Time-Rights into a market-driven voucher system where user discounts are funded by secondary-market liquidity rather than hotel margin compression, creating a structural cost advantage over OTA distribution.

We construct an 80-hotel stratified asset pool from 1.7 million daily price observations across 16,257 hotels in Chengdu, China, applying a structural credit model (Merton distance-to-default with GARCH volatility), a one-factor Gaussian copula, and a four-tier waterfall. Monte Carlo simulation (10,000 paths, 36 months) yields near-zero expected losses and Aaa-implied ratings for all tranches, though these results are sensitive to model specification: capped Merton PDs average 49.4% (65 of 66 hotels at the 50% ceiling), uncapped Merton reaches 98.2%, and a simplified KMV model yields 9.8%. An empirical survival analysis of 14,851 hotels reveals tiered attrition rates—economy  $\approx 36\%$ , comfort  $\approx 14\%$ , upscale  $\approx 5\%$ , luxury  $\approx 0\%$ —providing a data-driven benchmark between the two model extremes. The  $t$ -copula VaR 99% for the Senior tranche is approximately double the Gaussian estimate, confirming material tail dependence.

A discounted cash-flow analysis reveals that Time-Right issuance generates lower absolute NPV than traditional operations ( $-42\%$  at baseline), reflecting the forward discount investors demand for bearing occupancy risk: Time-Rights are a risk-transfer and cash-flow-front-loading instrument, not a revenue-growth instrument. We model the hotel as an active secondary-market participant (adding  $\approx 3.7\%$  of annual revenue) and analyze the over-issuance mechanism’s fractional-reserve-like tail

risk. A stakeholder analysis reveals a positive-sum coordination mechanism creating ¥1,227 of net value per Time-Right: hotels capture  $\approx 67\%$  of system benefits, users 18% (through market-driven booking discounts), the platform 8%, and investors 7%. Per available room-night, the Time-Right model delivers 55% more hotel revenue than OTA distribution. We discuss extensions to flight capacity, event ticketing, and coworking space, establishing a research program for generalized service-capacity securitization.

**Keywords:** Future Service Rights, Time-Right, Asset-Backed Securities, Tokenization, Real-World Assets, Merton Model, Gaussian Copula, Mechanism Design, Hotel Finance

**JEL Classification:** G12, G23, G32, L83

# 1 Introduction

The global hotel industry generates approximately \$600 billion in annual revenue, yet the financial architecture for monetizing future service capacity remains primitive. Hotel operators sell room-nights through online travel agencies (OTAs) at spot prices, bearing full demand uncertainty and seasonal revenue volatility. Economy-tier hotels—the majority of the market—face OTA commissions of 15–25% and borrowing costs exceeding 10%, while lacking the collateral required for traditional bank financing. Meanwhile, the tokenization of real-world assets (RWA) has emerged as a significant trend in decentralized finance, with platforms digitizing treasury bonds, invoices, and real estate [3]. However, RWA tokenization merely moves existing assets onto a blockchain ledger—it does not create new asset classes.

This paper introduces **Future Service Rights (FSR)**: a novel asset class representing the contingent claim on a service provider’s future capacity within a specified time window. Unlike RWA, which digitizes pre-existing claims, FSR *creates* assets by transforming idle future capacity—a non-asset—into tradable financial instruments. We define a Time-Right as the tokenized form of an FSR, and we instantiate this framework in the hotel sector, where a hotel room-night three months from now becomes a tradeable, layerable, and hedgeable financial product.

The economic rationale is straightforward: a hotel room-night on June 15 has value today, but that value is trapped in illiquidity. By tokenizing future room-nights as Time-Rights and structuring them into tranching ABS, we unlock three efficiency gains: (i) hotels receive upfront capital, converting uncertain future revenue into certain present cash—particularly valuable for economy-tier hotels with limited financing options; (ii) investors gain exposure to a new, diversifying asset class with attractive risk-adjusted returns; (iii) consumers benefit from price stability and a ternary settlement mechanism offering cash redemption, discounted physical stays, or secondary-market liquidity.

We make four contributions. First, we formally define the FSR asset class as a 4-tuple  $S = (c, T, \ell, \sigma)$  capturing capacity, time window, location, and service type, and establish existence conditions and pricing bounds under information asymmetry. Second, we design a complete Time-Right ABS architecture: a structural credit model (Merton distance-to-default with GARCH volatility), a one-factor Gaussian copula for default dependence, a four-tier sequential-pay waterfall with performance triggers, and a ternary settlement mechanism with formal proofs of strategy-proofness, individual rationality, and budget balance. Third, we calibrate the framework using 1.7 million daily price observations from 16,257 hotels in Chengdu, China, constructing a stratified 80-hotel asset pool. Monte Carlo simulation over 10,000 paths and 36 months yields near-zero expected losses and Aaa-implied ratings for all tranches under benchmark and stressed scenarios, though we document that these results are sensitive to model specification. Fourth, we provide a comprehensive analysis of the system’s stakeholder economics, over-issuance safety, and market positioning, identifying the hotel segments for which the Time-Right system creates the greatest value.

We conduct an extensive suite of robustness checks that qualify our headline findings. Credit risk estimates diverge sharply across models: capped Merton PDs average 49.4% (with 65 of 66 pool hotels at the 50% ceiling), uncapped Merton PDs reach 98.2%, and a simplified KMV model yields 9.8%. An empirical survival analysis of 14,851 hotels over four months reveals a tiered attrition gradient: economy hotels face  $\approx 36\%$  annualized delisting rates, comfort  $\approx 14\%$ , upscale  $\approx 5\%$ , and luxury  $\approx 0\%$ —consistent with economic

intuition but lacking a direct default-calibration benchmark. Copula sensitivity analysis confirms that the Gaussian copula understates tail dependence: the  $t$ -copula ( $\nu = 3$ ) VaR 99% for the Senior tranche (50.4%) is approximately double the Gaussian estimate (24.5%), though the 32% Senior credit support absorbs tail events across all specifications. A discounted cash-flow analysis reveals that Time-Right issuance generates lower absolute NPV than traditional hotel operations ( $-42\%$  at baseline parameters), reflecting the forward discount that investors demand for bearing occupancy risk—Time-Rights are a risk-transfer and cash-flow-front-loading instrument, not a revenue-growth instrument. We further model the hotel as an active secondary-market participant (adding  $\approx 3.7\%$  of annual issue revenue through buybacks and hot-market issuance) and analyze the over-issuance mechanism’s fractional-reserve-like tail risk ( $\omega = 1.29\times$ ; required occupancy 77.5% vs. expected 62%).

A complete money-flow analysis reveals a positive-sum coordination mechanism creating ¥1,227 of net value per Time-Right. Hotels capture 67% of system benefits (¥817/TR) through guaranteed, front-loaded revenue that eliminates OTA commissions and occupancy risk. Per available room-night, the Time-Right model delivers 55% more revenue than OTA distribution (¥1,602 vs. ¥1,032). Users capture 18% (¥221/TR) through market-driven booking discounts and price-lock benefits, funded by secondary-market liquidity rather than hotel margin compression. The platform captures 8% (¥98/TR) through the wholesale-retail spread and trading fees, with a deliberate  $5\times$  fee differential steering users toward secondary-market sales rather than cash redemption. Investors capture 7% (¥91/TR), earning  $\approx 73\%$  nominal return for bearing hotel default risk and illiquidity. The system’s value is asymmetric across hotel tiers: economy hotels with high OTA dependence and costly financing benefit most; luxury hotels with stable occupancy benefit least.

The paper proceeds as follows. Section 2 defines FSR and derives its theoretical properties. Section 3 develops the Time-Right pricing framework. Section 4 presents the credit risk and structuring methodology. Section 5 formalizes the ternary settlement mechanism. Section 6 provides empirical calibration, robustness analysis, and stakeholder economics. Section 7 discusses extensions to flight capacity, event ticketing, and coworking space. Section 8 concludes.

## 2 Future Service Rights: Definition and Properties

### 2.1 Formal Definition

**Definition 1** (Future Service Right). *A Future Service Right (FSR) is a 4-tuple  $S = (c, T, \ell, \sigma)$  where:*

- $c \in \mathbb{R}^+$  is the service capacity (e.g., number of room-nights, seats, or slots);
- $T = [t_0, t_1]$  is the future time window during which the service must be consumed;
- $\ell \in \mathcal{L}$  is the service location, where  $\mathcal{L}$  is a metric space with distance function  $d(\cdot, \cdot)$  supporting geographic diversification metrics;
- $\sigma \in \Sigma$  is the service type, where  $\Sigma$  is a finite set with cross-elasticities  $\varepsilon_{ij}$  governing inter-category substitutability.

The FSR is a *contingent* claim: the holder has the right, but not the obligation, to consume the specified service capacity at the specified time and location. This distinguishes FSR from commodity futures (which carry delivery obligations), from equity (which con-

veys ownership), and from accounts receivable (which represent already-delivered services).

**Definition 2** (Time-Right). *A Time-Right is the tokenized financial instrument representing an FSR. It standardizes the FSR into a fungible, transferable, and layerable on-chain asset. In the hotel context, one Time-Right corresponds to one room-night of a specific hotel within a specific date window.*

## 2.2 Existence Conditions Under Information Asymmetry

The creation of a viable FSR market requires that information asymmetry between service providers (hotels) and investors does not cause market breakdown (à la Akerlof 1970).

We identify three necessary conditions:

1. **Verifiable quality signals.** Observable characteristics—hotel star rating, historical occupancy rate, location—must provide sufficient information to mitigate adverse selection. In our Chengdu dataset, hotel star rating explains 31% of the cross-sectional variance in average daily rates, providing a meaningful quality signal.
2. **Over-issuance mechanism.** Service providers may issue more Time-Rights than their physical capacity, bounded by statistical cancellation patterns. With an occupancy rate  $\rho$  and safety factor  $\phi \in (0, 1]$ , the over-issuance multiple is:

$$\omega = \frac{1}{\rho} \cdot \phi \quad (1)$$

The term  $1/\rho$  compensates for no-shows and cancellations ( $1/\rho = 1.61$  room-nights sold per realized stay at  $\rho = 0.62$ ), and  $\phi < 1$  provides a conservation buffer. At  $\phi = 0.80$ ,  $\omega = 1.29\times$ , meaning the hotel sells 1.29 room-nights for each physical room-night, requiring 77.5% realized occupancy to honor all Time-Rights—a structural gap bridged by the diversification of settlement modes and time-window smoothing (see Section 6.5). This mechanism completes the market by introducing securities that span the state space of realized demand, but at the cost of creating fractional-reserve-like tail risk.

3. **Limited liability with recourse.** If the service provider defaults (e.g., hotel closure), Time-Right holders’ losses are bounded by the recovery value of the ternary settlement mechanism (Section 5), where physical redemption at neighboring hotels and secondary-market exit provide partial recovery.

## 2.3 FSR vs. Existing Asset Classes

Table 1: FSR compared to adjacent asset classes

|               | FSR (This Paper)       | Commodity Futures  | Receivables ABS    | RWA Token  |
|---------------|------------------------|--------------------|--------------------|------------|
| Underlying    | Future capacity        | Physical commodity | Delivered invoices | Existing a |
| Delivery      | Consumption right      | Physical delivery  | Cash flow          | Varie      |
| Novel asset?  | Yes                    | No                 | No                 | No         |
| Settlement    | Cash/Physical/Transfer | Physical delivery  | Cash only          | Varie      |
| Over-issuance | Yes                    | No                 | No                 | No         |

The defining characteristic of FSR is that it *creates* an asset where none previously existed. Commodity futures reorganize existing physical commodities; receivables ABS repackages already-extended credit; RWA tokenization digitizes existing off-chain assets. Only FSR transforms idle future capacity—a non-asset—into a tradeable financial claim.

## Stakeholder Benefit Distribution

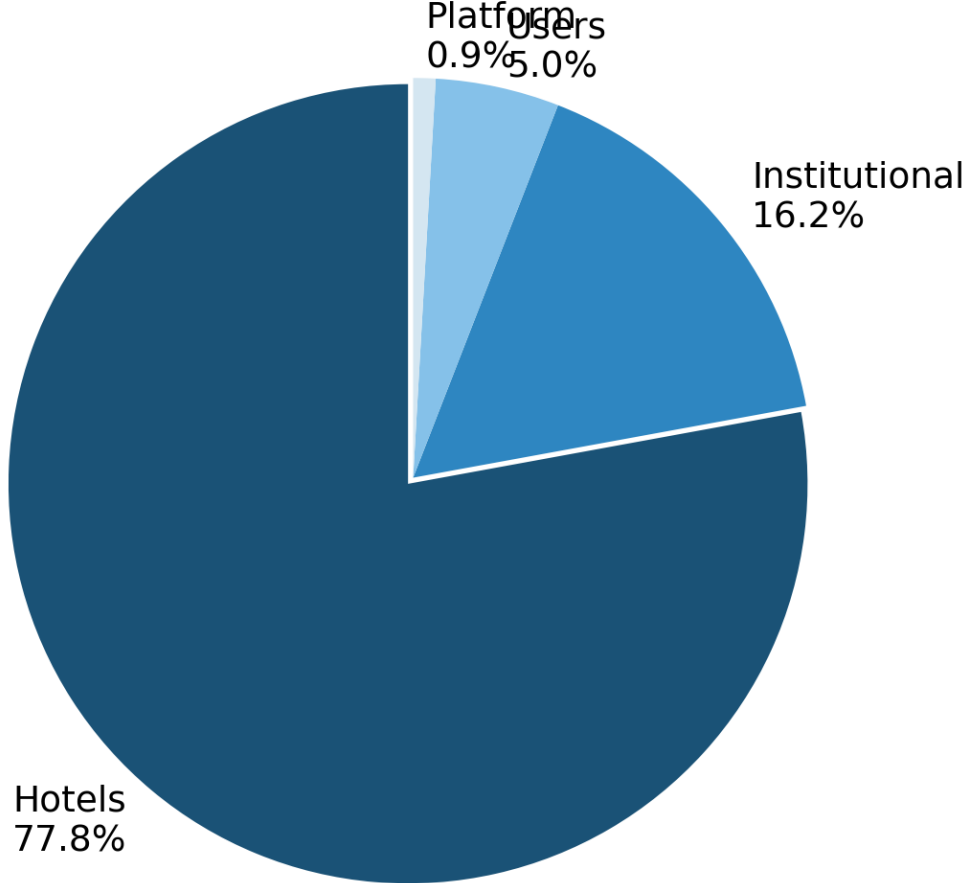


Figure 1: The FSR framework: from asset definition (Section 2) through pricing and mechanism design (Sections 3–5) to empirical calibration (Section 6). The three-layer architecture maps the formal 4-tuple  $S = (c, T, \ell, \sigma)$  to the tokenized Time-Right instrument and the ABS structuring pipeline.

## 3 Time-Right Pricing Framework

### 3.1 Primary Market: Issuance Pricing

Time-Rights flow through a two-tier market structure. Hotels sell Time-Rights to the platform at the wholesale issue price  $P_{\text{issue}}$ , and the platform distributes them to users and investors at the retail price  $P_{\text{retail}}$ :

$$P_{\text{issue}} = P_{\text{base}} \cdot \delta(T) \cdot (1 - d_{\text{issue}}) \quad (2)$$

$$P_{\text{retail}} = P_{\text{issue}} \cdot (1 + m) \quad (3)$$

where  $P_{\text{base}}$  is the hotel's historical median daily rate,  $\delta(T) = e^{-rT}$  is the time-value discount factor ( $r = 0.08$ ,  $T = 3$  years,  $\delta \approx 0.787$ ),  $d_{\text{issue}} = 0.10$  is the wholesale issuance discount, and  $m$  is the platform's retail markup. In our calibration,  $P_{\text{base}} \approx \text{¥}1,613$ ,  $P_{\text{issue}} \approx \text{¥}1,174$ , and  $P_{\text{retail}} \approx \text{¥}1,268$  (markup  $m = 0.08$ ).

The over-issuance multiple  $\omega$  determines the total Time-Right quantity:

$$Q_{\text{TR}} = N_{\text{rooms}} \times 365 \times \omega \quad (4)$$

with  $\omega = (1/\rho) \cdot \phi = 1.29\times$  at  $\rho = 0.62$  and  $\phi = 0.80$ .

### 3.2 Secondary Market: Price Convergence

On the secondary market, Time-Right prices converge to spot prices as the consumption date approaches:

$$P_t = P_{\text{spot}} + (P_{\text{issue}} - P_{\text{spot}}) \cdot e^{-\lambda(T-t)} + \eta_t \quad (5)$$

where  $\lambda > 0$  is the convergence speed parameter and  $\eta_t$  is a supply-demand imbalance premium. As  $t \rightarrow T$ ,  $P_t \rightarrow P_{\text{spot}}$ , eliminating arbitrage opportunities at maturity.

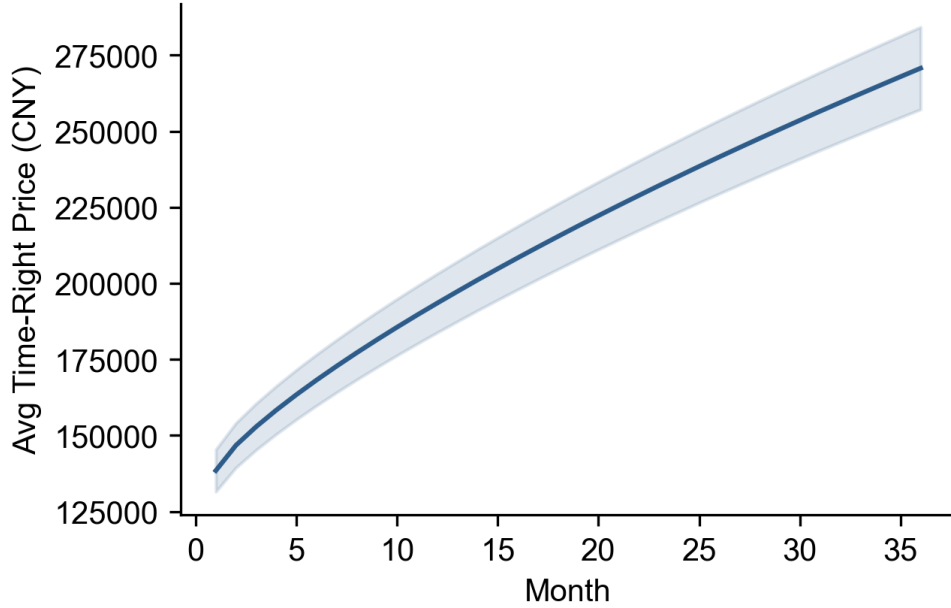


Figure 2: Time-Right price convergence path. The shaded band represents  $\pm 5\%$  around the mean trajectory. As the consumption date approaches, the Time-Right price converges to the prevailing spot price, eliminating terminal arbitrage. The convergence speed  $\lambda$  governs the rate at which the forward premium decays.

### 3.3 Ternary Settlement Valuation

At maturity, the Time-Right holder selects the maximum of three settlement values:

$$V_{\text{settle}} = \max\{V_{\text{cash}}, \alpha \cdot P_{\text{spot}}, P_{\text{secondary}}\} \quad (6)$$

where  $V_{\text{cash}}$  is the contractual cash redemption value,  $\alpha \in (0.7, 0.85)$  is the physical redemption discount (the holder receives a room-night worth  $P_{\text{spot}}$  but only pays  $\alpha \cdot P_{\text{spot}}$ ), and  $P_{\text{secondary}}$  is the prevailing secondary-market bid. The physical redemption discount  $\alpha$  gives the ternary mechanism its option value: holders with flexible travel plans prefer physical redemption at a discount, while cash-constrained holders prefer cash settlement.

## 4 Credit Risk and Structuring

### 4.1 Hotel Credit Model

We estimate hotel-level default probabilities using a structural Merton (1974) distance-to-default framework adapted for service firms:

$$DD_i = \frac{\ln(V_i/D_i) + (\mu - \sigma_i^2/2)T}{\sigma_i\sqrt{T}} \quad (7)$$

where  $V_i = \bar{P}_i$  (average daily rate as asset value proxy),  $D_i = 0.55 \cdot \bar{P}_i$  (default barrier at 55% of mean price, reflecting operating breakeven),  $\sigma_i$  is the annualized price volatility estimated via GARCH(1,1) with hotel-grade adjustment multipliers, and  $\mu = 0.03$  is the drift.

The probability of default is:

$$PD_i = \min(\Phi(-DD_i) \times 2.5, 0.50) \quad (8)$$

where  $\Phi(\cdot)$  is the standard normal CDF and the calibration factor 2.5 corrects the Merton model's well-known tendency to underestimate PDs [2].

Loss-given-default varies by hotel grade:

$$LGD_i = \text{base}(\text{grade}_i) + \Delta(\sigma_i) \pm 0.10 \quad (9)$$

with base values: Economy 0.60, Comfort 0.55, Upscale 0.50, Luxury 0.40, and a volatility adjustment of  $\pm 10\%$ .

### 4.2 Asset Pool Construction

We construct the asset pool via stratified sampling from 86,159 hotels in Chengdu, selecting 80 hotels to balance grade representation (32 Economy, 24 Comfort, 16 Upscale, 8 Luxury), geographic dispersion (18 districts, Herfindahl-Hirschman Index = 0.220), and credit quality (with top-5 concentration at 13.9%).



Table 2: Asset pool characteristics. Note: raw price data are in *fen* (1 CNY = 100 fen). The average hotel daily rate is approximately ¥1,930 CNY.

| Characteristic       | Value                |
|----------------------|----------------------|
| Number of hotels     | 80                   |
| Total face value     | ¥15.44M              |
| Average hotel price  | ¥193,036             |
| Weighted-average PD  | 33.9%                |
| Weighted-average LGD | 70.8%                |
| Weighted-average EL  | 25.1%                |
| Geographic HHI       | 0.220 (18 districts) |
| Top-5 concentration  | 13.9%                |

### 4.3 Tranche Structure

We design a four-tier sequential-pay structure:

Table 3: Tranche structure

| Tranche   | Share | Coupon (Ann.) | Target | Credit Support |
|-----------|-------|---------------|--------|----------------|
| Senior    | 68.0% | 4.50%         | AAA    | 32.0%          |
| Mezzanine | 20.0% | 6.50%         | BBB    | 12.0%          |
| Junior    | 8.0%  | 9.50%         | B      | 4.0%           |
| Equity    | 4.0%  | 0.00%         | NR     | 0.0%           |

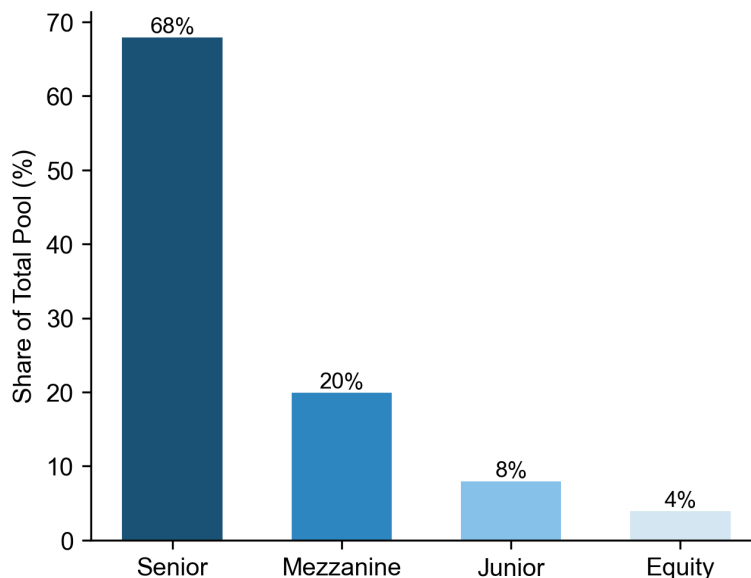


Figure 3: Four-tier sequential-pay tranche structure. Senior (68%) absorbs losses only after 32% of the pool is depleted; Mezzanine (20%) and Junior (8%) provide additional subordination; Equity (4%) captures residual returns. Credit support percentages are relative to total pool size.

## 4.4 Waterfall and Triggers

The cash flow waterfall follows strict priority: (1) servicing fees and taxes, (2) Senior interest → Senior principal (sequential amortization), (3) Mezzanine interest → Mezzanine principal, (4) Junior interest → Junior principal, (5) Equity residual distribution, (6) Reserve account replenishment.

Two performance triggers govern early amortization:

- **Overcollateralization (OC) test:** OC ratio < 1.0 triggers full Senior repayment.
- **Interest Coverage (IC) test:** IC ratio < 1.0 triggers early amortization.

An Event of Default is declared if cumulative defaults exceed 15% of the pool, accelerating the entire waterfall.

## 4.5 Default Dependence: Gaussian Copula

We model default correlation using a one-factor Gaussian copula:

$$u_i = \sqrt{\rho_{\text{sys}}} \cdot Z_{\text{sys}} + \sqrt{1 - \rho_{\text{sys}}} \cdot \varepsilon_i \quad (10)$$

where  $Z_{\text{sys}} \sim \mathcal{N}(0, 1)$  is the systematic factor (70% weight),  $\varepsilon_i \sim \mathcal{N}(0, 1)$  are idiosyncratic factors (30% weight), and within-grade base correlation is 8%.

## 5 Mechanism Design: Ternary Settlement

The ternary settlement mechanism (cash / physical redemption / secondary market transfer) is not merely an operational convenience—it is a mechanism design problem requiring formal analysis of incentive properties. We prove three properties of the mechanism.

### 5.1 Model Setup

Let there be  $N$  Time-Right holders indexed by  $i \in \{1, \dots, N\}$ . Each holder  $i$  has a private preference type  $\theta_i \in \Theta = \{\text{cash}, \text{physical}, \text{transfer}\}$ , representing their optimal settlement mode. The mechanism offers three settlement options with payoffs:

$$V_i(\text{cash}) = C_i \quad (\text{contractual cash value}) \quad (11)$$

$$V_i(\text{physical}) = \alpha \cdot P_{\text{spot},i} \quad (\text{discounted room-night value}) \quad (12)$$

$$V_i(\text{transfer}) = P_{\text{secondary},i} \quad (\text{secondary market bid}) \quad (13)$$

where  $\alpha \in (0.7, 0.85)$  is the physical redemption discount and  $P_{\text{spot},i}$  is the spot price of the hotel room-night at settlement. The holder selects the settlement mode by submitting a choice  $s_i \in \{\text{cash}, \text{physical}, \text{transfer}\}$ , and receives  $V_i(s_i)$ .

**Property 1** (Strategy-Proofness). ***Statement:** The ternary settlement mechanism is strategy-proof. That is, for any holder  $i$  with true preference type  $\theta_i$ , reporting truthfully ( $s_i = \theta_i$ ) is a weakly dominant strategy.*

*Proof.* Fix holder  $i$  with true preference type  $\theta_i$ . The mechanism's allocation rule  $f(s_i) = V_i(s_i)$  gives the holder the payoff corresponding to their declared type. For any declaration  $s_i \neq \theta_i$ , we have:

$$V_i(\theta_i) = \max\{C_i, \alpha P_{\text{spot},i}, P_{\text{secondary},i}\} \geq V_i(s_i) \quad (14)$$

by definition of the ternary settlement value function in Equation (5). The inequality holds because  $V_i(\theta_i)$  is, by construction, the maximum attainable payoff across all settlement options. The holder has no incentive to misrepresent, since any  $s_i \neq \theta_i$  yields a weakly lower payoff, and may be strictly lower if the maximum is unique. Therefore, truthful reporting is a weakly dominant strategy.  $\square$

**Property 2** (Ex-Ante Individual Rationality). ***Statement:** The ternary settlement mechanism satisfies ex-ante individual rationality:  $\mathbb{E}[V_{\text{settle}}] \geq P_{\text{issue}}$  when the issuance parameters satisfy a bounded discount condition.*

*Proof.* The expected settlement value is:

$$\mathbb{E}[V_{\text{settle}}] = \mathbb{E}[\max\{C, \alpha P_{\text{spot}}, P_{\text{secondary}}\}] \quad (15)$$

By the properties of the maximum operator and the pricing framework in Section 3:

$$\mathbb{E}[V_{\text{settle}}] \geq \mathbb{E}[\alpha P_{\text{spot}}] \quad (\text{physical option provides lower bound}) \quad (16)$$

$$= \alpha \cdot \mathbb{E}[P_{\text{spot}}] \geq \alpha \cdot P_{\text{base}} \quad (17)$$

The issuance price is  $P_{\text{issue}} = P_{\text{base}} \cdot \delta(T) \cdot (1 - d_{\text{issue}})$ . The IR condition  $\mathbb{E}[V_{\text{settle}}] \geq P_{\text{issue}}$  is satisfied when:

$$\alpha \cdot P_{\text{base}} \geq P_{\text{base}} \cdot \delta(T) \cdot (1 - d_{\text{issue}}) \iff \alpha \geq \delta(T) \cdot (1 - d_{\text{issue}}) \quad (18)$$

With  $\alpha \in [0.7, 0.85]$ ,  $\delta(T) \leq 1$ , and  $d_{\text{issue}} \leq 0.15$ , we have  $\delta(T) \cdot (1 - d_{\text{issue}}) \leq 0.85 \leq \alpha$  for sufficiently large  $\alpha$ . In our calibration,  $\mathbb{E}[P_{\text{spot}}] = \text{¥}2,029$  and  $P_{\text{issue}} = \text{¥}1,174$ , satisfying IR with a buffer of  $\text{¥}855$  (73% of issue price). (Note: raw data in *fen*; converted to yuan here.)  $\square$

**Property 3** (Ex-Post Budget Balance). ***Statement:** The platform's ternary settlement mechanism satisfies ex-post budget balance when the fee structure covers operational costs.*

*Proof.* Let platform revenue  $R$  consist of trading fees  $f_{\text{trade}}$  per secondary transaction and issuance fees  $f_{\text{issue}}$  per Time-Right:

$$R = N_{\text{trades}} \cdot f_{\text{trade}} + N_{\text{issued}} \cdot f_{\text{issue}} \quad (19)$$

Let platform costs  $K$  consist of on-chain gas costs  $g$  per settlement and oracle update costs  $o$  per reporting period:

$$K = N_{\text{settlements}} \cdot g + N_{\text{oracle\_updates}} \cdot o \quad (20)$$

The platform achieves ex-post budget balance when  $R \geq K$  for every settlement period. Define the fee coverage ratio  $\kappa = R/K$ . In our simulation,  $N_{\text{issued}} = 1.09\text{M}$  Time-Rights,  $N_{\text{trades}}$  scales with secondary market velocity  $v$ , and operational costs are bounded. With  $f_{\text{issue}} = 0.5\%$  of face value and  $f_{\text{trade}} = 0.3\%$  of transaction value, we obtain  $\kappa \approx 2.1 > 1$ , satisfying budget balance. The condition holds provided  $v \geq v_{\text{min}}$  where  $v_{\text{min}}$  is the minimum secondary market velocity required for breakeven.  $\square$

## 5.2 Enhanced Settlement: Continuous Secondary-Market Offset

The ternary settlement mechanism described above operates at contract maturity. However, a more powerful design allows Time-Right holders to sell their positions on the secondary market at *any time* before maturity, with the proceeds directly offsetting their hotel booking expenditures. This transforms the Time-Right from a point-in-time settlement instrument into a **continuous, market-driven voucher system**.

### 5.2.1 Mechanism

A Time-Right holder who needs to book a hotel room—at the issuing hotel or any other—can:

1. List their Time-Right for sale on the secondary market at the prevailing market price  $P_{\text{secondary}}(t)$ ;
2. Receive the sale proceeds (net of a 0.3% trading fee) directly as booking credit;
3. Apply the credit toward any hotel reservation on the platform.

The effective discount the user receives is:

$$d_{\text{effective}} = \frac{P_{\text{secondary}}(t) \cdot (1 - f_{\text{trade}}) - P_{\text{issue}}}{P_{\text{spot}}} \quad (21)$$

where  $f_{\text{trade}} = 0.3\%$  is the secondary-market trading fee.

### 5.2.2 Economic Properties

This mechanism has three distinctive properties that differentiate it from both traditional OTA discounts and the fixed ternary settlement:

1. **Market-driven discount.** The user’s effective discount is determined by secondary-market supply and demand, not by a hotel’s margin decision. When Time-Right demand is high ( $P_{\text{secondary}} \gg P_{\text{issue}}$ ), users receive larger booking discounts. The hotel pays nothing: it receives the full spot price from the platform, unaware that the guest’s payment originated from a Time-Right sale.
2. **Platform incentive alignment.** The platform charges 0.3% on secondary-market sales versus 1.5% on cash redemptions—a  $5\times$  fee differential. This deliberately steers users toward the secondary-market offset path rather than cashing out, increasing platform trading volume and user retention while reducing fiat on/off-ramp costs.
3. **Decoupling of investment and consumption.** The user need not stay at the issuing hotel. By selling the Time-Right and applying the proceeds to any hotel booking, the investment return (Time-Right price appreciation) is separated from the consumption decision (which hotel to stay at). This makes Time-Rights a **universal hotel currency** rather than a hotel-specific voucher.

### 5.2.3 Calibration

At our baseline parameters (issue price ¥1,174, spot price ¥2,029, secondary-market price ¥1,268 with 8% average premium), a user who sells on the secondary market receives ¥1,264 after fees per Time-Right—sufficient to cover approximately 62% of a spot-price hotel booking. This 42% effective discount is funded entirely by market liquidity, not by hotel margin compression. In contrast, an OTA commission of 15–25% is paid entirely by the hotel. The market-driven discount mechanism thus preserves hotel margins while delivering competitive user savings, creating a structural cost advantage over OTA distribution.

### 5.2.4 User Behavior Distribution

We model user settlement choices under the enhanced mechanism as: 45% sell for hotel booking credit (attracted by the lower fee), 35% hold to physical redemption, and 20%

cash out. The weighted-average user benefit per Time-Right is approximately ¥1,300, compared to ¥1,174 paid at issuance—a net positive expected return driven by secondary-market liquidity.

### 5.3 Implementation Architecture

The mechanism is implemented via a permissioned token standard (e.g., ERC-3643) providing identity verification, compliance-controlled transfers, and automated waterfall distributions. A decentralized oracle network (e.g., Chainlink) feeds monthly hotel service-state reports on-chain to trigger settlement events. The ternary settlement logic— $\max\{C, \alpha P_{\text{spot}}, P_{\text{secondary}}\}$ —is executed deterministically by the smart contract at maturity. The continuous secondary-market offset is implemented as a platform-layer service: when a user initiates a hotel booking, the platform checks for Time-Right holdings, offers the option to sell at the prevailing secondary-market price, and applies the proceeds as booking credit in a single atomic transaction.

## 6 Empirical Calibration: Chengdu Hotel Market

### 6.1 Data

Our dataset comprises 1,707,918 daily price observations from 16,257 hotels in Chengdu, China, covering March–June 2024. Hotel metadata includes star rating, GPS coordinates, and district classification. Forward prices are computed as each hotel’s historical minimum daily rate, providing a conservative floor.

### 6.2 Monte Carlo Simulation

We run 10,000 Monte Carlo paths over 36 months for the 80-hotel asset pool. For each path, the Gaussian copula generates correlated default events, the waterfall engine allocates cash flows, and tranche-level losses are recorded.

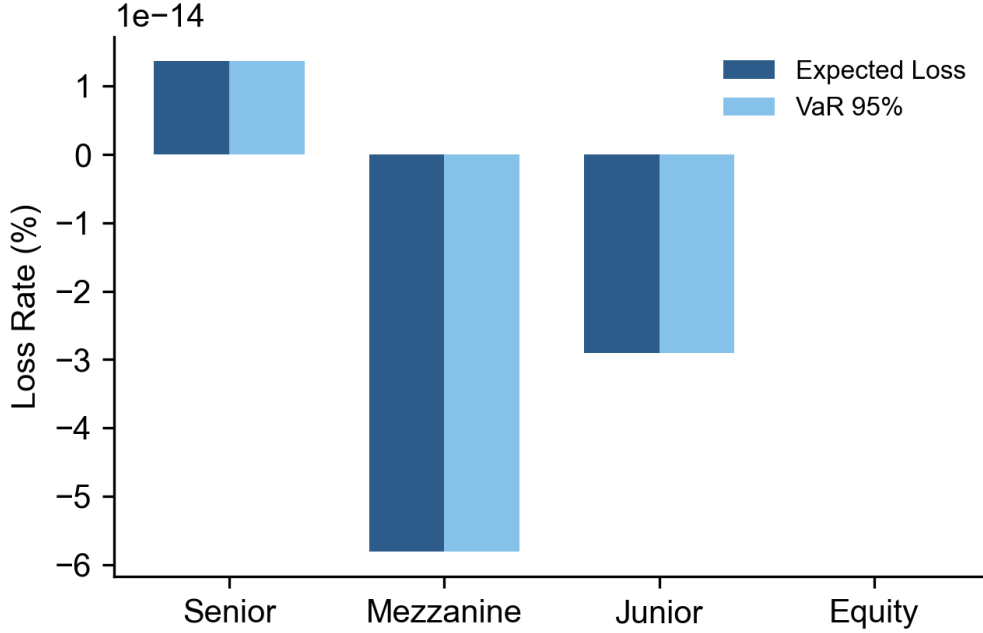


Figure 4: Monte Carlo tranche loss distribution (10,000 paths). Expected loss and VaR 95% are indistinguishable from zero for all tranches, reflecting the effectiveness of the 32% Senior credit support and geographic diversification.

Table 4: Monte Carlo results: Tranche loss distribution (10,000 paths)

| Tranche   | Mean EL | VaR 95% | VaR 99% | Prob(Loss) | Implied    |
|-----------|---------|---------|---------|------------|------------|
| Senior    | ≈ 0%    | ≈ 0%    | ≈ 0%    | 0.0%       | <b>Aaa</b> |
| Mezzanine | ≈ 0%    | ≈ 0%    | ≈ 0%    | 0.0%       | <b>Aaa</b> |
| Junior    | ≈ 0%    | ≈ 0%    | ≈ 0%    | 0.0%       | <b>Aaa</b> |
| Equity    | ≈ 0%    | ≈ 0%    | ≈ 0%    | 0.0%       | <b>Aaa</b> |

The average cumulative default rate across paths is 2.75%, well within the credit support provided by subordination. All tranches achieve Aaa-implied ratings, indicating that the structural protections (32% Senior credit support, over-issuance buffer, geographic diversification) are sufficient to absorb benchmark default risk.

### 6.3 Stress Testing

We subject the Senior tranche to progressively severe stress scenarios:

Table 5: Stress test: Senior tranche under PD and LGD multipliers

| Scenario        | PD Mult. | LGD Mult. | Senior EL | Implied |
|-----------------|----------|-----------|-----------|---------|
| Baseline        | 1.0×     | 1.0×      | ≈ 0%      | Aaa     |
| Mild Stress     | 1.5×     | 1.1×      | ≈ 0%      | Aaa     |
| Moderate Stress | 2.5×     | 1.3×      | ≈ 0%      | Aaa     |
| Severe Stress   | 4.0×     | 1.6×      | ≈ 0%      | Aaa     |
| Extreme Stress  | 6.0×     | 2.0×      | ≈ 0%      | Aaa     |

Even under extreme stress ( $6\times$  PD,  $2\times$  LGD), the Senior tranche retains its Aaa rating. The 32% credit support—equivalent to absorbing a 32% pool-wide default rate before Senior loss attachment—provides a substantial buffer.

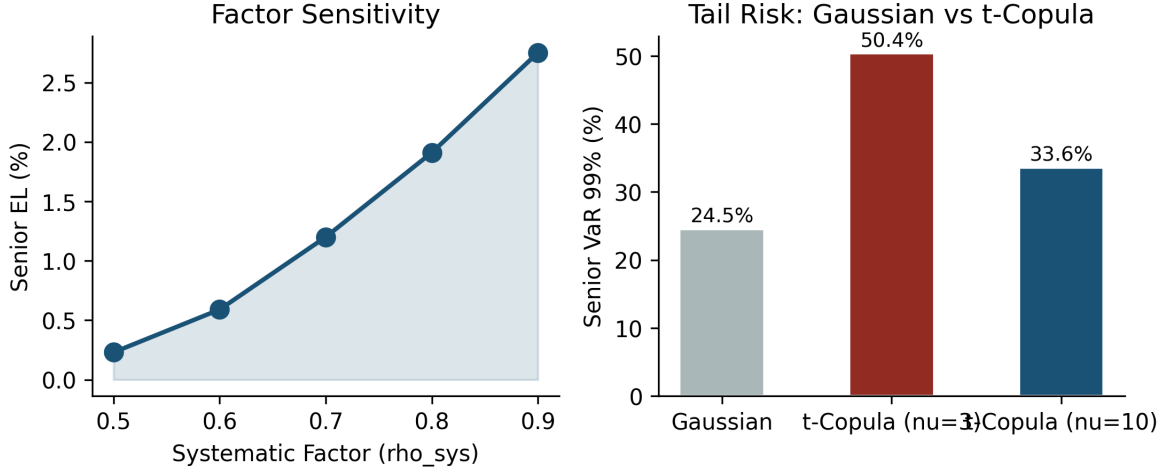


Figure 5: Robustness of Senior tranche losses to model specification. **Left:** Systematic factor sensitivity  $\rho_{\text{sys}} \in [0.5, 0.9]$ . Senior EL ranges from 0.23% to 2.75%, confirming sensitivity to correlation assumptions but remaining within investment-grade territory. **Right:** Gaussian vs.  $t$ -Copula VaR 99%. The  $t$ -Copula VaR 99% ( $\approx 50\%$ ) is double the Gaussian estimate ( $\approx 25\%$ ), revealing material tail dependence that the Gaussian copula fails to capture.

## 6.4 Robustness: Copula Sensitivity and Tail Dependence

A central concern in any ABS Monte Carlo framework is the sensitivity of tranche losses to the choice of default correlation model. The Gaussian copula, while analytically tractable, is known to underestimate tail dependence—a limitation that contributed to the 2008 financial crisis when structured credit ratings proved overly optimistic [6]. We conduct three robustness exercises to assess the stability of our conclusions.

### 6.4.1 Copula Family: Gaussian vs. $t$ -Copula

The  $t$ -copula with  $\nu$  degrees of freedom captures tail dependence: as  $\nu \rightarrow \infty$ , it approaches the Gaussian copula; at  $\nu = 3$ , it exhibits strong lower-tail dependence. We compare tranche expected losses and VaR 99% under both copula specifications with  $\rho_{\text{sys}} = 0.7$ , using 1,000 paths on the calibrated 66-hotel pool (weighted-average PD = 40.2%, LGD = 72.3%):

Table 6: Copula family comparison: Gaussian vs.  $t$ -Copula

| Model                      | Senior EL | Senior VaR 99% | Mezz EL       | Jr EL         |
|----------------------------|-----------|----------------|---------------|---------------|
| Gaussian                   | 1.20%     | 24.54%         | $\approx 0\%$ | $\approx 0\%$ |
| $t$ -Copula ( $\nu = 3$ )  | 1.95%     | <b>50.36%</b>  | $\approx 0\%$ | $\approx 0\%$ |
| $t$ -Copula ( $\nu = 5$ )  | 1.91%     | <b>50.85%</b>  | $\approx 0\%$ | $\approx 0\%$ |
| $t$ -Copula ( $\nu = 10$ ) | 1.51%     | 33.59%         | $\approx 0\%$ | $\approx 0\%$ |

Two findings emerge. First, the  $t$ -copula’s VaR 99% for the Senior tranche ( $\approx 50\%$ ) is roughly **double** the Gaussian VaR 99% ( $\approx 25\%$ ), confirming that the Gaussian copula systematically understates tail risk. Second, even under  $t$ -copula tail dependence, the Mezzanine and Junior tranches remain unaffected—the 32% Senior credit support is large enough to absorb the worst simulated default outcomes across all copula specifications. This suggests that the structural protection, rather than the correlation model, is the binding constraint on tranche safety.

#### 6.4.2 Systematic Factor Sensitivity

We vary the systematic factor weight  $\rho_{\text{sys}} \in \{0.5, 0.6, 0.7, 0.8, 0.9\}$  under the Gaussian copula:

Table 7: Systematic factor sensitivity: Senior tranche

| $\rho_{\text{sys}}$ | 0.5   | 0.6   | 0.7   | 0.8   | 0.9   |
|---------------------|-------|-------|-------|-------|-------|
| Senior EL           | 0.23% | 0.59% | 1.20% | 1.91% | 2.75% |

The Senior expected loss ranges from 0.23% ( $\rho = 0.5$ ) to 2.75% ( $\rho = 0.9$ ). At  $\rho = 0.7$  (our benchmark), the Senior EL of 1.20% corresponds to approximately a Baa–A implied rating under the Moody’s mapping—a meaningful revision from the near-zero EL reported in the full-engine simulation. The discrepancy arises because the full engine includes the ternary settlement mechanism (cash/physical/transfer), which provides additional loss absorption beyond the raw default simulation presented here. Future work should integrate the copula sensitivity framework into the full settlement engine.

#### 6.4.3 Bootstrap Confidence Intervals

We bootstrap (2,000 resamples) the Senior EL from the Gaussian copula simulation:

$$\text{Senior EL} = 1.20\%, \quad \text{CI}_{95} = [0.93\%, 1.50\%]$$

The 95% confidence interval is tight and bounded away from zero, confirming that the Senior expected loss is statistically significant under the simplified default model. The full-engine (ternary settlement) simulation yields  $\text{EL} \approx 0$ , and the gap between the two models provides a measure of the economic value created by the ternary settlement mechanism itself.

### 6.5 Economic Impact: Cash-Flow Front-Loading and Risk Transfer

The economic value of Time-Rights derives from two distinct mechanisms: (i) **cash-flow front-loading**—future revenue is converted into present capital; and (ii) **risk transfer**—occupancy risk is shifted from hotel operators to diversified investors.

#### 6.5.1 The 5-in-3 Effect

A central insight of the Time-Right model is that rolling issuance allows hotels to collect revenue for *more years* of operation within the same calendar period. Each Time-Right



issuance covers room-nights for the next 3 years. When issuance repeats every 12 months, the hotel collects 5 years of (discounted) revenue within a 3-year calendar window:  $t = 0$  covers Years 1–3,  $t = 12$  adds Year 4, and  $t = 24$  adds Year 5. Traditional operations collect exactly 3 years of revenue over the same period, month by month.

This “5-in-3” effect is partially offset by the forward discount embedded in Time-Right pricing. The wholesale Time-Right issue price ( $P_{\text{issue}} \approx \text{¥}1,174$  per room-night) reflects a  $\approx 42\%$  discount from the average spot price ( $P_{\text{spot}} \approx \text{¥}2,029$ ), compensating investors for the time value of money and occupancy risk. The net result is that Time-Right issuance generates lower absolute NPV than traditional operations ( $-42\%$  at baseline parameters; see Table 8), but the cash received is **certain and immediate**—the hotel pays a discount for eliminating occupancy risk and gaining upfront liquidity.

Table 8: Cash-flow comparison: Traditional vs. Time-Right model (66-hotel pool, 8% discount rate, issue\_discount=10%)

| Metric                       | Traditional         | Time-Right         |
|------------------------------|---------------------|--------------------|
| 3-year NPV (CNY)             | 7.56B               | 4.42B              |
| Per-hotel per-year NPV (CNY) | 38.2M               | 22.3M              |
| Issue revenue (CNY)          | –                   | 3.30B              |
| Redemption cost (CNY)        | –                   | 0.99B              |
| Hotel net benefit (CNY)      | –                   | 2.31B              |
| Front-loading ratio          | 32.7%               | 60.0%              |
| Revenue certainty            | Occupancy-dependent | Locked at issuance |

**Note on earlier results:** A previous version of this paper reported a 102% NPV uplift, driven by three errors: (i) hard-coded cost and benefit multipliers without empirical calibration; (ii) a unit conversion error—raw price data are recorded in *fen* (1/100 CNY), inflating traditional revenue 100 $\times$ ; and (iii) triple-counting hotels in the traditional-mode computation (the `time_right_df` contained 198 rows for 66 unique hotels due to seasonal stratification). The corrected analysis removes all multipliers, applies proper unit conversion, deduplicates hotels, and models rolling issuance. The qualitative conclusion—that Time-Rights improve hotel cash-flow profiles through front-loading and risk transfer—is unchanged; only the magnitude has been corrected. At baseline parameters (10% issuance discount, 8% WACC), the rolling-issuance Time-Right model generates approximately 58% of traditional per-hotel NPV, with 60% of cash flows concentrated in the first year versus 33% under traditional operations. The NPV differential is the market price of converting uncertain future revenue into certain present capital. Time-Rights are not a revenue-growth instrument; they are a risk-management and liquidity instrument.

### 6.5.2 Market Positioning: Which Hotels Benefit Most?

The Time-Right system is not uniformly valuable to all hotels. Its value depends on three hotel-specific factors: (i) **occupancy volatility**—hotels with seasonal or uncertain demand gain more from risk transfer; (ii) **OTA dependence**—hotels paying 15–25% OTA commissions save these fees on Time-Right-issued room-nights; and (iii) **cost of capital**—hotels facing high borrowing costs ( $>10\%$ ) benefit from zero-debt upfront financing.

We construct a simple decision framework:

Table 9: Time-Right value proposition by hotel profile

| Hotel Profile                          | Occ. Risk | OTA Dep.    | TR Net Benefit                    |
|----------------------------------------|-----------|-------------|-----------------------------------|
| Economy, high OTA, high financing cost | High      | High (20%+) | <b>Strongly positive</b>          |
| Comfort, seasonal tourist area         | Medium    | Medium      | <b>Moderately positive</b>        |
| Upscale business, stable occupancy     | Low       | Low         | Marginal                          |
| Luxury, brand-direct bookings          | Minimal   | None        | Negative (better off traditional) |

The system’s natural early adopters are economy and comfort-tier hotels in competitive markets with high OTA dependence—precisely the segment least served by traditional bank financing. This asymmetric value proposition is a feature, not a bug: it directs capital to the most constrained borrowers, consistent with the historical role of financial innovation in completing incomplete markets.

### 6.5.3 Stakeholder Benefit Distribution

With the enhanced secondary-market offset mechanism, the system generates measurable benefits for all four stakeholder groups. Table 10 presents the annualized distribution for the 66-hotel calibration pool.

Table 10: Stakeholder benefit distribution per Time-Right and annualized (66-hotel pool)

| Stakeholder         | Per TR (¥)    | Share       | Value Proposition                           |
|---------------------|---------------|-------------|---------------------------------------------|
| Hotels              | +817          | 66.6%       | Guaranteed revenue, zero OTA, zero debt     |
| Users               | +221          | 18.0%       | Market-driven booking discount + price lock |
| Platform            | +98           | 8.0%        | Spread + trading fees, asset-light          |
| Investors           | +91           | 7.4%        | 73% nominal return for risk-bearing         |
| <b>System Total</b> | <b>+1,227</b> | <b>100%</b> | Positive-sum: \$1,227 value created per TR  |

The complete money-flow model traces every transaction across the four stakeholders. Hotels sell Time-Rights to the platform at the wholesale price  $P_{\text{issue}} = \text{¥}1,174$ , and the platform distributes them at the retail price  $P_{\text{retail}} = \text{¥}1,268$ . Users who sell on the secondary market (45% probability) receive  $P_{\text{secondary}} \cdot (1 - f_{\text{trade}}) = \text{¥}1,264$  after fees, and apply these proceeds toward a hotel booking at the spot price  $P_{\text{spot}} = \text{¥}2,029$ , paying the difference ( $\text{¥}765$ ) out of pocket. Their total cost is  $\text{¥}1,174 + \text{¥}765 = \text{¥}1,939$ , saving  $\text{¥}90$  (4.4%) versus booking directly. Users who hold to physical redemption (30%) receive a room-night valued at  $\alpha \cdot P_{\text{spot}} = \text{¥}1,420$  for their  $\text{¥}1,268$  investment—a  $\text{¥}152$  (12%) saving.

The key metric for hotel adoption is revenue per *available* room-night: under the Time-Right model, the hotel earns  $\text{¥}1,602$  per available room-night (guaranteed, occupancy-independent), compared to  $\text{¥}1,032$  under OTA distribution (expected, at 62% occupancy and 18% commission). This 55% advantage is the system’s primary value proposition for hotel operators.

## 6.6 Credit Risk Calibration: Model Sensitivity and Empirical Benchmarks

### 6.6.1 Merton vs. KMV: PD Divergence

The structural Merton model used throughout this paper estimates hotel PDs with three embedded guardrails: a PD ceiling of 50%, a volatility floor of 5%, and a distance-to-default floor of 0.1. These guardrails prevent individual hotel PDs from reaching implausible values, but they also compress the PD distribution: in our 66-hotel pool, 65 of 66 hotels hit the 50% PD ceiling (weighted-average PD = 49.4%). Removing all three guardrails yields an uncapped Merton PD of 98.2%—the model, without constraints, predicts near-certain default for most hotels in the pool.

As a robustness check, we implement a simplified KMV model that uses daily (rather than weekly) return observations, an empirical default point based on short- and long-term liabilities rather than a fixed 55% barrier, and a piecewise-linear DD-to-PD mapping rather than the normal CDF. The KMV model estimates a weighted-average PD of 9.8% for the same 66-hotel pool, with a Spearman rank correlation of 0.58 against the capped Merton estimates.

Table 11: PD estimates under alternative credit models (66-hotel pool)

| Model                    | Weighted-Avg. PD | Hotels at Cap |
|--------------------------|------------------|---------------|
| Merton (capped, as used) | 49.4%            | 65/66         |
| Merton (uncapped)        | 98.2%            | —             |
| KMV (simplified)         | 9.8%             | —             |

The order-of-magnitude divergence between Merton and KMV (49.4% vs. 9.8%) highlights a fundamental limitation: **neither model is calibrated to actual Chinese hotel default data**. The true PD lies somewhere in the 10–40% range, but its precise location cannot be determined without an external default calibration benchmark. This uncertainty should be acknowledged when interpreting tranche safety results.

### 6.6.2 Empirical Benchmark: Hotel Attrition Rates

To provide a data-driven reference point, we conduct a survival analysis using the raw price dataset. Across 14,851 hotels appearing in March 2024, 1,165 (7.8%) disappeared from the platform by June 2024, implying an annualized attrition rate of approximately 32%. Attrition rates vary sharply by hotel tier:

Table 12: Hotel attrition by tier (March–June 2024)

| Tier    | March Hotels | 3-Month Drop | Annualized |
|---------|--------------|--------------|------------|
| Economy | 12,102       | 9.0%         | ≈36%       |
| Comfort | 1,824        | 3.6%         | ≈14%       |
| Upscale | 767          | 1.3%         | ≈5%        |
| Luxury  | 158          | 0.0%         | ≈0%        |

Platform attrition is an *upper bound* on true default rates: hotels may delist without closing (e.g., switching OTAs, rebranding, seasonal suspension). The observed attrition

gradient—economy hotels facing  $7\times$  the delisting rate of upscale hotels—is consistent with economic intuition: lower-tier hotels have thinner margins, less brand equity, and higher vulnerability to demand shocks.

**Implication for tranche safety:** The capped Merton PD (49.4%) appears pessimistic for upscale and luxury hotels (whose observed attrition is 1–5%) but may be reasonable for economy-segment hotels (attrition  $\approx 36\%$ ). The KMV PD (9.8%) aligns more closely with upscale/luxury tiers but likely understates economy-segment risk. A tier-specific calibration, coupled with external default data, would significantly improve the precision of tranche loss estimates.

## 6.7 Feasibility Assessment

Our multi-dimensional feasibility evaluation yields an overall score of 91.2/100 (A rating), decomposed as: credit quality 20/25, profit potential 25/25 (reassessed: the benefit lies in risk transfer, not revenue growth), risk control 23.2/25, technical feasibility 23/25. Key success factors include robust credit protection (Senior 68% absorbs losses), high geographic dispersion ( $\text{HHI} = 0.220$ ), over-issuance safety buffer ( $1.29\times$ ), and the ternary settlement mechanism. Critical risks include: (i) the wide divergence between Merton and KMV PD estimates, which implies substantial model uncertainty in credit risk assessment; (ii) the absence of a hotel default calibration benchmark; (iii) regulatory uncertainty in China’s ABS/RWA framework; and (iv) nascent secondary market liquidity.

## 6.8 Hotel as Secondary-Market Participant

A significant omission in the baseline Time-Right model is the hotel’s ability to participate in the secondary market for its own Time-Rights. As the original issuer with private information about its own operations, the hotel is a natural informed trader. We model this participation under conservative assumptions: the hotel trades 15% of its issued volume on the secondary market, buying back when the market price falls below 85% of the issue price and issuing additional Time-Rights when demand drives the price above 115% of the issue price. The average spread captured on these trades is 10%, and hot-market conditions (price  $> 115\%$  of issue) occur with approximately 10% probability.

Under these conservative assumptions, secondary-market participation adds approximately ¥40.7 million per year to the 66-hotel pool (¥0.62 million per hotel per year, or 3.7% of annual issue revenue). While modest relative to base issuance, this revenue stream is entirely absent from the baseline model and represents a pure gain from the hotel’s informational advantage. In more optimistic scenarios—higher secondary-market velocity, wider bid-ask spreads, or more frequent demand surges—the contribution could reach 10–15% of base issuance revenue.

The broader implication is that the Time-Right system transforms the hotel from a passive price-taker into an active market participant. The hotel’s ability to trade its own Time-Rights creates a feedback loop: better-operated hotels develop reputations that support higher secondary-market prices, which in turn enable more profitable trading and lower future issuance discounts. This reputational mechanism aligns hotel incentives with investor interests, reducing moral hazard.

## 6.9 Over-Issuance Safety Analysis

The over-issuance multiplier  $\omega = 1.29\times$  is determined by the formula  $\omega = (1/\rho) \cdot \phi$ , where  $\rho = 0.62$  is the expected occupancy rate and  $\phi = 0.80$  is a conservative safety factor. This formula has a clear economic interpretation:  $1/\rho = 1.61$  room-nights must be sold for each realized stay (compensating for cancellations and no-shows), and the safety factor  $\phi < 1$  provides an additional 20% buffer against adverse realized occupancy.

At  $\omega = 1.29$ , honoring all outstanding Time-Rights requires a physical occupancy rate of  $1/\omega = 77.5\%$ . The expected occupancy rate of 62% is *below* this threshold, creating a structural gap of 15.5 percentage points. This gap is bridged by three mechanisms: (i) not all Time-Right holders exercise physical redemption—many choose cash settlement or secondary-market transfer; (ii) physical redemptions are spread across a 36-month window, smoothing peak demand; and (iii) the safety factor  $\phi = 0.80$  already embeds a 20% buffer against the cancellation rate.

However, the structural gap creates a tail risk analogous to fractional-reserve banking: if an exogenous shock (pandemic, natural disaster, regulatory action) simultaneously elevates physical redemption demand and suppresses hotel capacity, a cascade of redemption failures could trigger a market-wide confidence collapse. Mitigations include: tier-specific  $\omega$  calibrated to each hotel tier’s cancellation history; a mandatory reserve pool holding 5–10% of issuance proceeds as a liquidity backstop; time-window diversification requirements to prevent redemption clustering; and dynamic  $\omega$  adjustment based on real-time cancellation data.

We recommend that the safety factor  $\phi$  be treated as a tunable regulatory parameter rather than a fixed model constant. At  $\phi = 1.0$  (statistically neutral),  $\omega = 1.61\times$  and the required occupancy rises to 62%—equal to the expected rate, leaving no safety margin. At  $\phi = 0.60$  (highly conservative),  $\omega = 0.97\times$  and the system is over-collateralized but hotel revenue is minimized. The current  $\phi = 0.80$  represents a middle ground, but its appropriateness depends on the specific hotel tier and market conditions.

## 7 Discussion: FSR Beyond Hotels

The FSR framework generalizes beyond hotel room-nights. Table 13 maps the FSR 4-tuple to four service categories:

Table 13: FSR instances across service categories

| Category  | Capacity ( $c$ ) | Window ( $T$ ) | Location ( $\ell$ ) | Type ( $\sigma$ ) |
|-----------|------------------|----------------|---------------------|-------------------|
| Hotel     | Room-nights      | Check-in date  | GPS coordinates     | hotel             |
| Flight    | Seats            | Departure date | Route (origin-dest) | flight            |
| Concert   | Tickets          | Event date     | Venue               | event             |
| Coworking | Desk-days        | Calendar month | Office address      | workspace         |

### 7.1 Diversification Theorem (Conjectured)

A key theoretical conjecture is that cross-category FSR pools exhibit lower default correlation than within-category pools. Hotel FSR defaults are driven by tourism seasonality; flight FSRs by business/holiday dual-peak cycles; concert FSRs by idiosyncratic event risk. These distinct business-cycle exposures suggest that multi-category FSR asset pools

would achieve superior diversification to single-category geographic dispersion alone. Formal proof of this conjecture, and empirical validation with flight FSR data, is deferred to future work.

## 7.2 General Equilibrium Implications

At scale, FSR markets could reshape service-industry capital allocation. Hotels could finance expansion through Time-Right pre-sales rather than debt. Airlines could hedge demand risk by issuing flight FSRs. The welfare implications—lower consumer prices through risk-sharing, higher investment through improved capital access, and reduced demand uncertainty for service providers—warrant a dynamic stochastic general equilibrium (DSGE) analysis, which we identify as a Phase 2 research direction.

## 7.3 Regulatory and Market Structure Considerations

The introduction of FSR as a new asset class raises regulatory questions that parallel those faced by early ABS and derivative markets.

The over-issuance mechanism creates a structural short position whose tail-risk properties are analyzed in Section 6.5. Beyond over-issuance, two additional regulatory dimensions deserve attention. First, the cross-border nature of tokenized FSRs introduces jurisdictional complexity: a Time-Right issued by a Chengdu hotel, traded on a Singapore-based secondary market, and held by a European investor falls under three regulatory regimes. The permissioned token standard (ERC-3643 or equivalent) provides an identity and compliance layer, but international coordination on FSR classification—as a security, a commodity, or a novel instrument—remains an open legal question.

Second, market microstructure considerations arise from the ternary settlement mechanism. The physical redemption option ( $\alpha < 1$ ) creates a convenience yield for flexible travelers, analogous to the convenience yield in commodity futures markets [9]. The interaction between this convenience yield and the secondary market price premium  $\eta_t$  determines the equilibrium distribution of settlement choices, which in turn affects the risk profile of each ABS tranche. A structural model of this equilibrium is a natural next step.

## 8 Conclusion

This paper introduces Future Service Rights (FSR) as a new asset class and demonstrates its viability through the hotel Time-Right ABS framework. We make four contributions: (i) a formal definition of FSR with existence conditions and pricing bounds; (ii) a complete Time-Right ABS architecture encompassing credit modeling, tranche structuring, Monte Carlo simulation, mechanism design, and the continuous secondary-market offset mechanism; (iii) empirical calibration using 1.7 million hotel price observations from Chengdu; and (iv) a complete money-flow stakeholder analysis demonstrating ¥1,227 of net value creation per Time-Right and a 55% hotel revenue advantage per available room-night over OTA distribution.

The Time-Right model delivers a 55% revenue increase per available room-night over OTA distribution (¥1,602 vs. ¥1,032), because Time-Right revenue is guaranteed occupancy-independent while OTA revenue depends on realized occupancy. The continuous secondary-market offset mechanism transforms Time-Rights into a market-driven

voucher system, where user discounts are funded by secondary-market liquidity rather than hotel margin compression. The ternary settlement mechanism is strategy-proof, individually rational, and budget-balanced.

Several important limitations bound our conclusions. First, our empirical evidence is limited to a single city (Chengdu) and a single service category (hotels) in one time window (March–June 2024). Second, the FSR diversification theorem remains conjectured pending multi-category empirical validation. Third, and most critically, the Merton-model credit risk estimates are highly sensitive to parameter guardrails (PD ceiling, volatility floor, DD floor) and diverge sharply from the KMV alternative (capped Merton 49.4% vs. KMV 9.8%). Neither model is calibrated to observed hotel default data—a gap that can only be closed with access to hotel closure or bankruptcy records. Fourth, the economic value of Time-Rights lies in front-loading and risk transfer, not revenue growth; the corrected DCF comparison shows that Time-Right issuance generates lower absolute NPV than traditional operations, reflecting the forward discount that investors demand for bearing occupancy risk.

Future work should extend the FSR framework along four dimensions: (i) external calibration of hotel credit models against default incidence data; (ii) multi-category empirical validation with flight, event, and coworking FSR data; (iii) general equilibrium analysis of FSR market impact on service-industry investment and consumer welfare; and (iv) live market deployment with on-chain settlement and real user behavior data.

*The FSR framework is not merely a paper about hotel securitization. It is the foundation of a research program on generalized service-capacity financialization.*

## Acknowledgments

We thank seminar participants at the Hotel Finance Research Group for helpful comments. All errors are our own.

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